

PROOF OF CONCEPT · 2026

Native PWA Experiment

A synthetic case showing that a progressive web app today can be indistinguishable from a native mobile or desktop application — including the UI itself.

Installable

Offline

Per-platform native UI



THE IDEA

**whatpwacando.today catalogs what the web *can* do.
This experiment shows how it can **feel**.**

1

One synthetic app

"Streaks", a daily habit tracker — small enough to read, real enough to judge.

2

Native UI per platform

Cupertino on iOS & macOS, Material on Android & Linux, Fluent on Windows.

3

Judged like a native app

Install flow, offline start, dark mode, safe areas, platform widgets and gestures.

1

codebase — domain logic shared by every UI

4

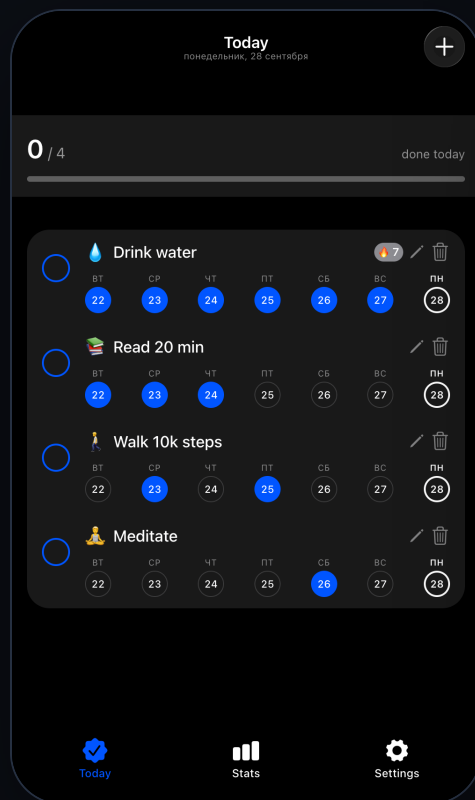
native design systems from one tree

0

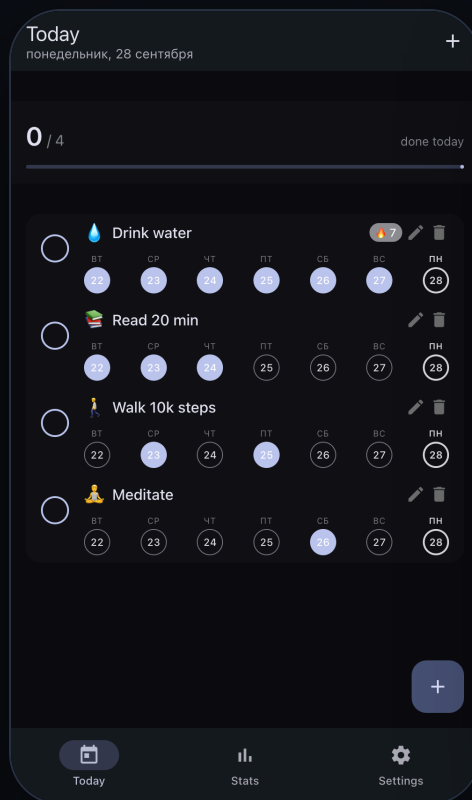
app store reviews to ship an update

Streaks — the same app, four native looks

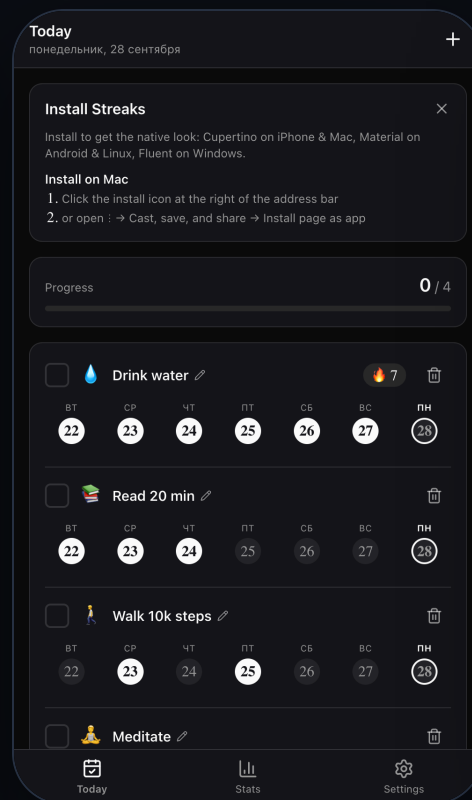
The UI kit is chosen at runtime: detected for installed PWAs, forceable via `?design=...` for previews.



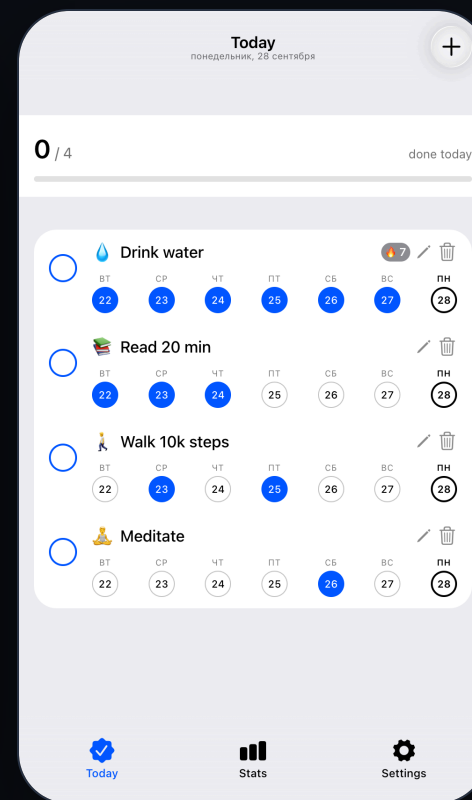
Cupertino · iOS / macOS



Material · Android / Linux



Web kit · browser tab



Light & dark · every kit

One domain layer, swappable UI kits

Domain — src/domain

Pure reducer, versioned localStorage persistence, streak & progress selectors. Zero UI imports, fully unit-tested (80 tests).

Platform — src/platform

Detects OS, browser, display mode and install status; resolves the design system (override → installed → web) and owns the install prompt store.

PWA shell

vite-plugin-pwa: manifest, auto-updating service worker, per-kit lazy chunks, offline precache.



Cupertino · Material

Framework7 9 — one shell, two native languages (`theme="ios" / "md"`), F7 icons, swipe actions, FAB, system dialogs.

Fluent

@fluentui/react-components v9 — real Fluent widgets, CSS-in-JS design tokens, TabList navigation, dark theme from the OS.

Web

Own Flutter-style widget kit — inline styles from a single theme file (`theme.jsx`), no CSS frameworks at all.

Details that close the gap

01

Installed-mode detection

display-mode queries + navigator.standalone switch the whole design system the moment the app leaves the browser tab.

02

Real install flow

beforeinstallprompt is captured and re-fired; where the prompt is missing (iOS), per-browser instructions are shown.

03

Offline by default

the service worker precaches the build — cold start works in airplane mode.

04

Platform widgets & gestures

FAB and pill tabs on Android, circular checks and swipe-to-delete on iOS, TabList on Windows — from one component tree.

05

Native chrome

safe-area insets for notches and home bars, system dark mode, per-platform typography and icon fonts.

06

Lean bundles

each kit ships as a lazy chunk: a browser tab never downloads Framework7, an installed Android user never downloads Fluent.

PWA vs native binary — what's left on the table?

	THIS EXPERIMENT (PWA)	NATIVE BINARY
Distribution	A URL — no store, no review	App store submission & review
Updates	Instant, on next load (autoUpdate SW)	Release cycle + user update
UI quality	Platform UI kits per OS	Platform UI kits
Offline	Precached app shell + data	Full offline
Deep OS access	Moderate — web APIs only	Full platform access
Codebase	One for every platform	Per-platform (Swift/Kotlin/...)

Honest scope: a PWA is not a replacement for apps that need deep OS integration — it is a serious answer for everything else.

Not the first to bet on the web

What PWA Can Do Today

The capability catalog this experiment pairs with: what browsers support, per platform. whatpwacando.today

Excalidraw · Squoosh

Flagship installable PWAs — app-grade UX fully in the browser, offline-capable.

Starbucks · Uber · X Lite

Production PWAs at scale — lighter, faster to market, near-parity with their native apps.

Framework7 · Fluent UI

The kits that make platform-native looks possible from web code.

web.dev/learn/pwa · PWA Stats

Reference learning path and outcome statistics from the field.

Run it in a minute

```
git clone
https://github.com/nikita-fedorov008/NativePWAExperiment
cd NativePWAExperiment
npm install
npm run dev    # http://localhost:5173
npm run build  # static dist/ – host anywhere
npm test      # 80 unit tests
```

Open it in a browser tab for the web look — then install it and watch the UI switch to your platform's native kit.

```
src/
├─ domain/ habits · celebration – pure logic + tests
├─ platform/ detection · install store
├─ app/ adaptive switcher
└─ ui/
    ├─ f7/ + cupertino/ + material/
    ├─ fluent/
    ├─ web/ – code-styled kit + theme
    └─ appStyles.js – all custom CSS in one file
```

TRY IT YOURSELF

Judge it on your own phone.

1

Open the URL

in Safari or Chrome

2

Tap Install

from the banner or browser menu

3

Launch it

the native UI of your platform greets you

github.com/nikita-fedorov008/NativePWAExperiment